

Pendulum smoothing and prediction

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Use the 600-second mean as the general rate estimate. It has no clock-family or impulse-cycle setting, preserves elapsed-time meaning, and gives a particularly steady output. Keep the established PPS calibration separate. Add a repeating-pattern predictor only if predicting individual swings is a useful requirement.

This recording covers 66.38 hours, with 119,448 eligible PPS-calibrated full swings. The first 24 hours establish the phase template, hours 24-42 select candidate settings, and the final 24.38 hours provide subsequent evaluation. No swing observations from the faulty 67-day run enter this analysis.

Ex-ante: forecasting the next complete 30-second cycle

Model	Next-cycle period RMSE (us)	Output change per cycle, RMS (us)	Next-hour elapsed RMSE (ms)
600-second mean	2.53	0.168	7.16
Dual EWMA (5/60 min half-lives)	4.22	0.136	10.79
600-second raw median	131.82	0.469	236.92
Median of 20 complete-cycle means	2.56	0.235	7.19
Phase-aware EWMA (120 s half-life)	2.58	0.397	6.88
Phase-aware local linear (1,800 s)	2.53	0.194	6.98

The mean, cycle-median, phase EWMA and phase linear model are effectively close for this decision: next-cycle RMSE is 2.53-2.58 us. Differences change rank over six-hour portions of the final day. The 600-second mean has the smallest output movement among those candidates. No claim of statistically significant superiority is made.

The long EWMA is smoother still, but its slower response costs accuracy here. Stability is useful only alongside bias and responsiveness. The raw median is biased by the asymmetric impulse pattern, so its robustness does not make it a suitable rate estimator for this record.

Ex-post: how close was the pendulum?

Mean full period: **2.000018772 seconds**. Against a nominal 2-second full swing, the equivalent clock rate is **-0.811 seconds/day** (losing time). The eligible observed intervals account for -2.242 seconds of nominal clock gain, or 2.242 seconds lost. This sum excludes unavailable intervals and is not a continuous phase claim across the recording gap.

Strengths and weaknesses

Option	Strengths	Weaknesses and appropriate use
600-second mean	Simple, general, direct elapsed-time interpretation. Good forecast accuracy and stable output. No repeating-cycle setting.	About 300 s of average age. A real change takes time to propagate. An isolated bad observation affects the mean until it leaves the window. Preferred general default.
Dual EWMA	Constant memory, very smooth, incremental. Useful short and long views. Already established for PPS.	The swing blend retains a long tail and more lag. It does not explicitly remove a strong periodic impulse pattern. Better as an optional slow trend view than as the primary rate model here.
Raw 600-second median	Resists isolated extreme observations. General, with a clear typical-observation meaning.	A typical swing need not represent elapsed time per swing. Here its next-cycle bias is about -131 us, equivalent to roughly +5.7 s/day of rate error. Do not use raw median for clock rate without proving negligible mean-median bias.
Median of complete-cycle means	Robustness after averaging legitimate impulse structure. Its forecasts are close to the mean in this run.	Requires a valid cycle definition. Less steady here (0.235 vs 0.168 us RMS output change). No observed accuracy benefit to justify a second default model.
Phase-aware EWMA	Balances drift tracking and noise reduction. Good next-cycle and longer forecast results. Constant-size state.	Requires reliable phase assignment and a representative pattern. Output moves more than the 600-second mean. A fixed template ages as the waveform changes.
Adaptive phase + mean	Best tested individual-swing forecast. Zero-mean pattern leaves the generic mean rate estimate intact.	Additional per-phase state and a cycle-length assumption. Individual phase noise and drift must be managed. Useful only for a specific prediction requirement.
Local linear / polynomial	Can reduce lag while a trend persists. Useful retrospective reconstruction.	Extrapolation can amplify noise, overshoot after a step, or project a trend that ends. Generic raw linear fit performs poorly here. Little gain from the phase-aware version over the simpler mean.

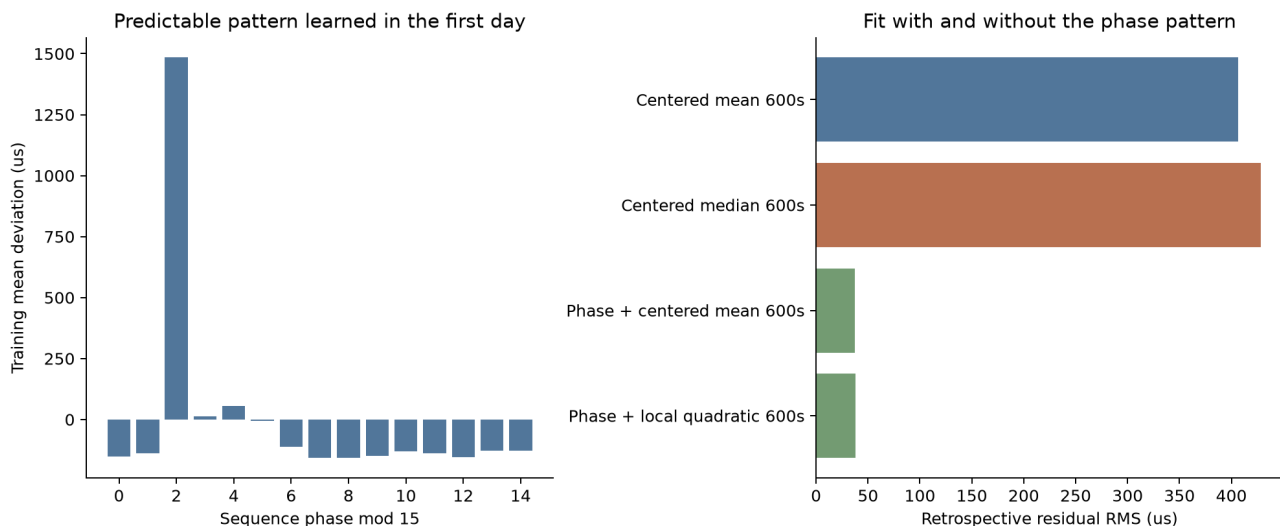
A local-level Kalman filter adds little as a separate candidate when it has fixed gains: its estimate follows the same constant-gain update as an EWMA. A trend-state Kalman model addresses similar trade-offs to the local linear candidate, but adds process/noise assumptions. This record does not show a benefit that warrants those extra settings. Neither Kalman candidate was independently implemented or benchmarked.

What is predictable about each swing?

The first-day phase profile spans 1642 us across the 15 sequence phases. This is legitimate repeating swing structure, not scatter that a median should automatically reject. A generic rate estimate predicts the average; an individual-swing forecast also needs the pattern.

Individual next-swing forecast	Final-period RMSE (us)
Previous swing	568.2
600-second mean	407.8
Previous swing at the same phase	27.8
Fixed first-day profile + phase-aware EWMA	37.8
Adaptive profile + 600-second mean	19.1

The adaptive model learns a zero-mean deviation for each sequence phase, with a 600-second half-life for shape updates. It predicts the next phase while holding the observed mean and shape fixed at the forecast origin. The shape adapts because a first-day profile alone transfers less well to the final day. Physical direction and the release event are not assigned by sequence phase.



The retrospective chart uses all observations available in the smoothing window. Its residuals describe agreement with observed swings; they do not measure error against an independently known true instantaneous rate. A lower residual from a shorter or more flexible smoother can simply mean closer fitting.

Generality without excessive settings

The default should not require a Synchronome setting. Use a trailing window measured in captured seconds, qualified PPS calibration, and a mean of completed full-swing durations. This is applicable to a different nominal pendulum period and to clocks with a more uniform impulse. Do not hard-code 300 swings: that approximates 600 seconds only for this 2-second pendulum.

The essential distinctions are output purpose, not clock brand: a rate estimate, a forecast made before a future swing, and a retrospective reconstruction. Keep those meanings explicit. A raw median can be retained as a diagnostic of the distribution without making it a second operating mode.

If individual-swing prediction is required, make the repeating-cycle length a single optional setting. A value of 1 disables a multi-phase waveform. The shape should have zero mean so it cannot redefine the rate. The current benchmark establishes only length 15. Before adopting another length, require chronological evidence that it improves forecasts. Avoid adding automatic cycle discovery, several filter selectors, and separate short/long/shape tuning controls before they solve a demonstrated need.

Illustrative uniform-impulse sensitivity check

These simulations retain this recording's slow drift, remove its deterministic impulse pattern, and add independent 20 us Gaussian noise with a fixed seed. A second scenario adds a +1 ms disturbance each hour. They test dependence on impulse shape; they are not physical evidence for another pendulum. The clean future mean is known only inside the simulation.

Model	Clean-rate forecast RMSE (us)	With disturbances RMSE (us)	Output change RMS (us)
Mean 600 s	2.09	2.48	0.38
Median 600 s	2.28	2.28	0.52
Dual EWMA 5/60 min	2.81	2.86	0.28
EWMA 120 s	2.19	2.88	0.88

Under symmetric uniform behaviour the median loses its large Synchronome bias and becomes competitive. With declared disturbances it is slightly more robust than the mean. The ordinary mean remains simpler to interpret and steadier in the clean scenario. This supports a generic mean default with retained anomaly evidence, rather than a universal median or a mandatory phase model.

Smallest useful design

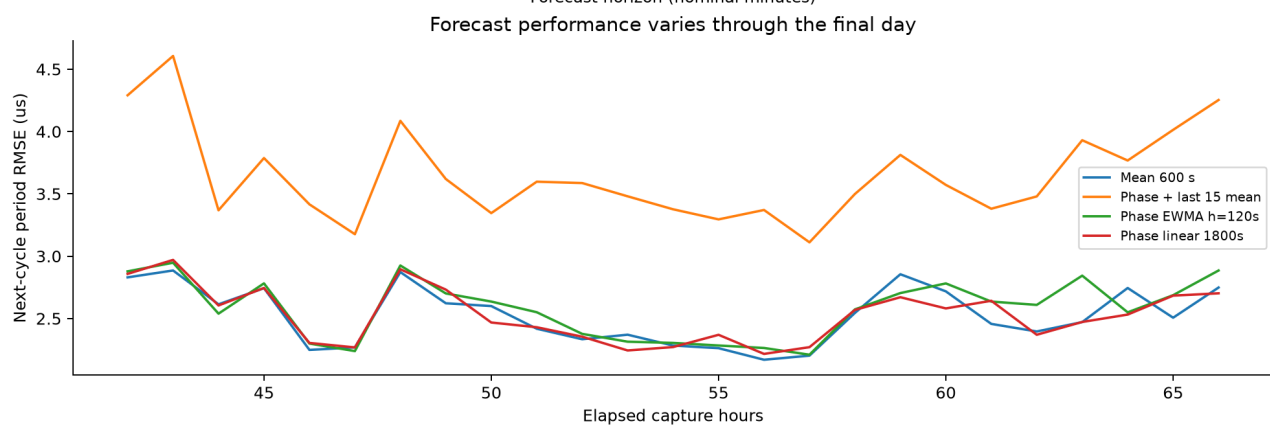
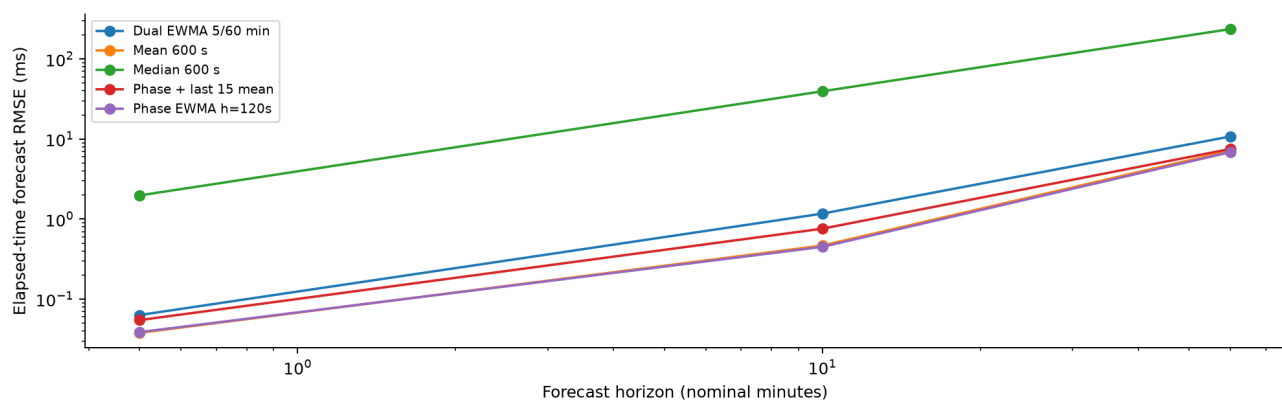
Keep PPS calibration as the reference layer. Use one generic swing-rate estimate, the 600-second mean. Add a clearly labelled ex-ante waveform forecast only when needed. Calculate ex-post rate and accumulated error directly from retained observations. This recommendation does not require changing firmware or the current capture path.

Stability, response and future horizons

Controlled sensitivity: a constant-mean repetition of the training waveform receives either one +1,000 us observation or a permanent +20 us mean-period step. The table reports estimator response, not observed pendulum behaviour. Step settling means remaining within 10% of the final shift.

Model	Peak effect of one 1 ms swing (us)	Step settling (minutes)	Step overshoot (us)
Mean 600 s	3.33	9.0	0.00
Dual EWMA 5/60 min	3.55	79.3	0.00
EWMA 60 min	0.39	199.3	0.00
Median 600 s	0.00	8.5	0.00
Phase EWMA h=120s	11.49	6.6	0.00
Phase linear 1800s	4.44	28.5	6.86
Median of 20 cycle means	0.00	5.5	0.00

The dual blend takes about 79 minutes to settle to a permanent mean-period change because the slow component retains 25% weight. The 600-second mean settles within about 9 minutes in this test. Local linear extrapolation overshoots. The faster phase EWMA responds sooner but is more sensitive to one anomalous swing. Median robustness applies to sparse disturbances; it does not establish that a repeated physical impulse is erroneous.



Future horizons contain 15, 300 or 1,800 consecutive full swings. Their nominal durations are 30 s, 10 min and 1 hour. Longer forecast windows overlap; origin counts are not independent experiments. Six-hour and hourly breakdowns are descriptive checks of changing performance.

Retrospective reconstruction and evidence

Retrospective measurement answers a different question: how much observed pendulum time accumulated against PPS? Sum the calibrated durations, then compare that elapsed time with the nominal indication of 2 seconds per completed full swing. Do not integrate a median or repeatedly smoothed display estimate as the authoritative accumulated clock error.

The longest calibrated continuous span is 42.888 hours, sequences 50-77,247. It loses 1.500847 seconds, equivalent to -0.840 s/day. A PPS-fit exclusion separates it from the later span. The sum over all eligible intervals loses 2.242317 seconds over 66.361 observed hours; unavailable periods are not filled.

Whole-record individual full-period scatter is 404.5 us. Complete 15-swing mean scatter is 14.65 us, including slow drift. Neither is a pure measurement-noise estimate.

Using hindsight without scoring a model against itself

Retrospective model, evaluated cycle excluded	Cycle-mean residual RMS (us)	Evaluated cycles
Neighbor-cycle mean 630s	2.42	2,895
Neighbor-cycle median 630s	2.45	2,895
Neighbor-cycle quadratic 630s	2.48	2,895
Neighbor-cycle mean 1830s	2.40	2,835
Neighbor-cycle median 1830s	2.41	2,835
Neighbor-cycle quadratic 1830s	2.39	2,835

These estimates use complete cycles on both sides but omit the evaluated cycle itself. Mean, median and quadratic fits are very close. About 2.4 us RMS remains between a cycle mean and its neighbouring reconstruction. That includes genuine cycle variation and metrology noise; the record does not identify their separate contributions. A centred mean of cycle averages is adequate for visualising slow rate changes.

Reference calibration and provenance

Forecast inputs replay the current causal PPS clock: 20 s and 3,600 s half-lives blended 75/25, with its qualified-edge gating. The current 600-second mean comparison instead divides mean swing ticks by mean qualified PPS ticks, matching the display implementation. The other models smooth causally calibrated swing durations. Retrospective targets use centred 61 s quadratic PPS phase regression and inverse-frequency integration. The RMS difference between dual-PPS and retrospective calibrated full-swing durations is 0.0217 us, far below the swing-model differences. This supports retaining the PPS approach here; it does not independently measure absolute PPS accuracy.

A separate 600-second swing mean using the dual-PPS input gives next-cycle RMSE 2.530 us versus 2.531 us for the current ratio-of-means implementation. The swing recommendation therefore does not depend materially on replacing the established PPS smoother.

Unsigned 32-bit counter wrap, sequence continuity, shared-counter alignment and calibration coverage are retained. Forecast learning uses raw-valid completed swings and causal calibration; a gap confined to retrospective calibration does not clear causal filter history. No global clipping, sorting, gap interpolation or raw-record edits are used. Recomputed complete-cycle means match the retained analysis export to below 0.0001 us.

Source revision: 4049212201494796b3a4310772f53a2ef04bfce4. The source checkout was clean. Exact input and implementation SHA-256 identities, settings, result CSVs and the executable analysis are retained in model_evaluation. This is an exploratory comparison on one record, with no independent second pendulum or long-run replication.